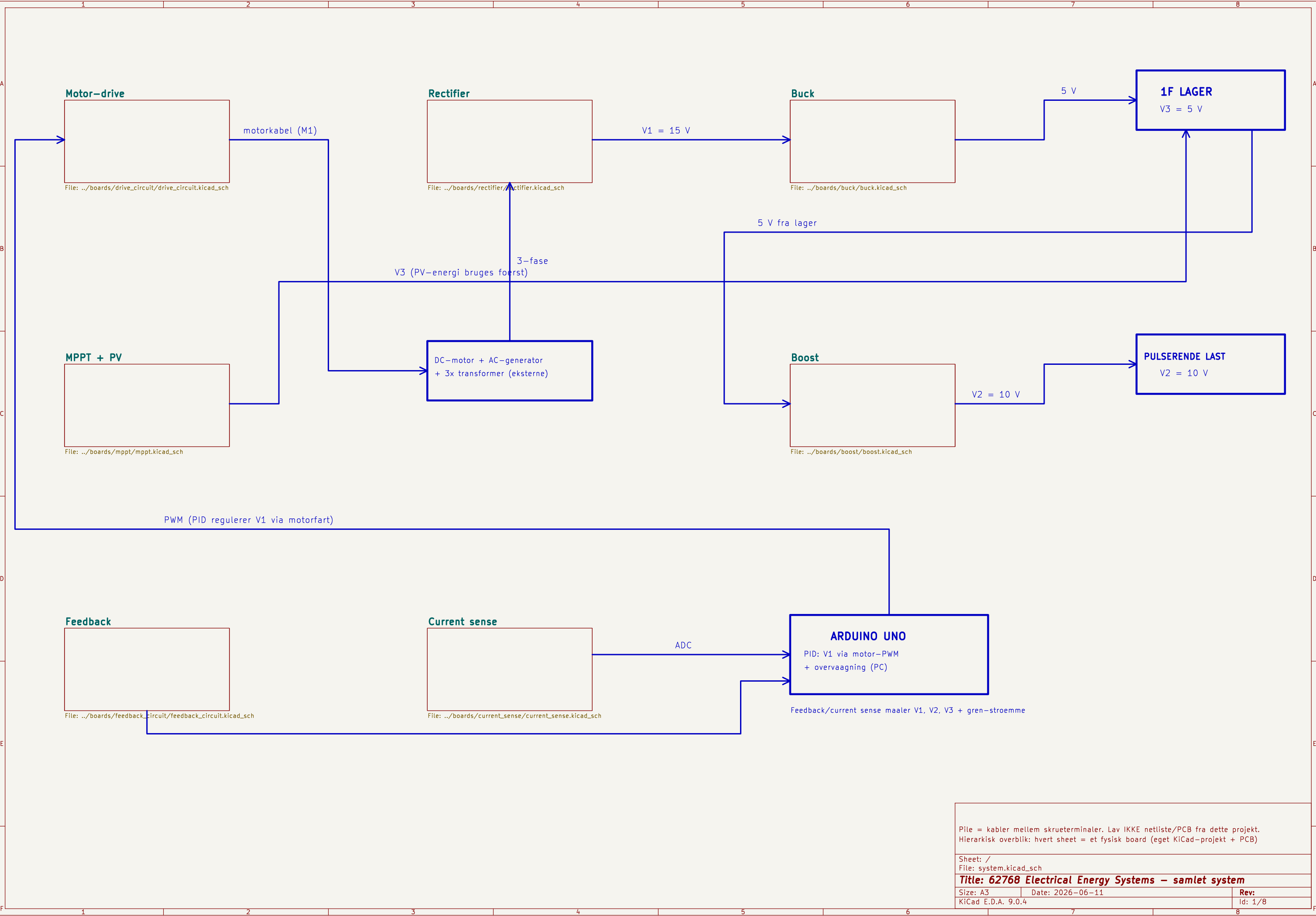




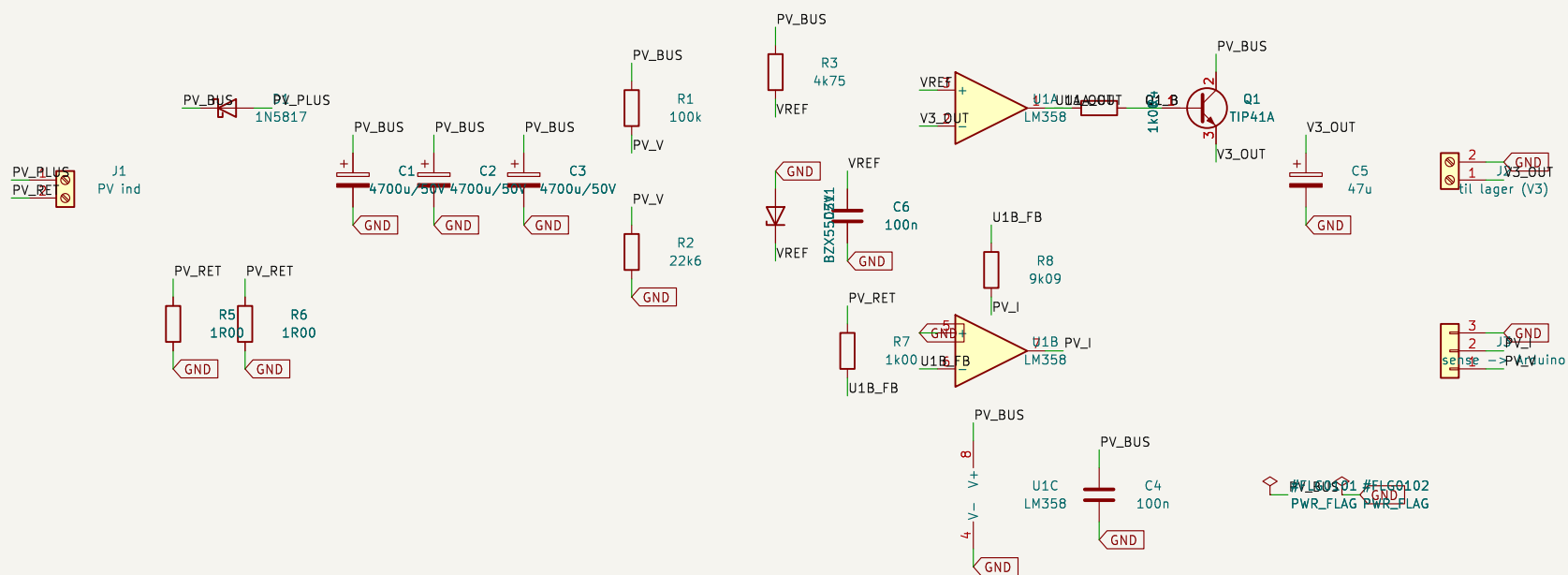
Drive Circuit (Exp 3A: PWM -> opto -> IR2110 -> IRL540N -> DC motor)

The circuit diagram illustrates a PWM motor drive system. It begins with a PWM signal input (J2) connected to a 200 ohm resistor (R1) and then to the input of an ILD74 optocoupler (U1A). The optocoupler's output is connected to the input of an IR2110 half-bridge driver (U2). The IR2110 is powered by a +15V supply with decoupling capacitors C1 (22uF) and C2 (100nF). The output of the IR2110 drives the gate of an IRL540N MOSFET (Q1) through a 10 ohm resistor (R2). The MOSFET's source is grounded, and its drain is connected to the motor (M1) through a 1k ohm resistor (R4). The motor is also protected by a 1N4007 diode (D1) in parallel. The motor is powered by a +20V supply. The circuit is labeled 'Drive Circuit (Exp 3A: PWM -> opto -> IR2110 -> IRL540N -> DC motor)'.

Sheet: /Motor-drive/		
File: drive_circuit.kicad_sch		
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Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.4	Id: 2/8	

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Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.4		Id: 2/8

PV front-end (PV -> 1N5817 -> ~15 mF -> diskret 5V-regulator -> lager + sense)



Sheet: /MPPT + PV/
File: mppt.kicad_sch

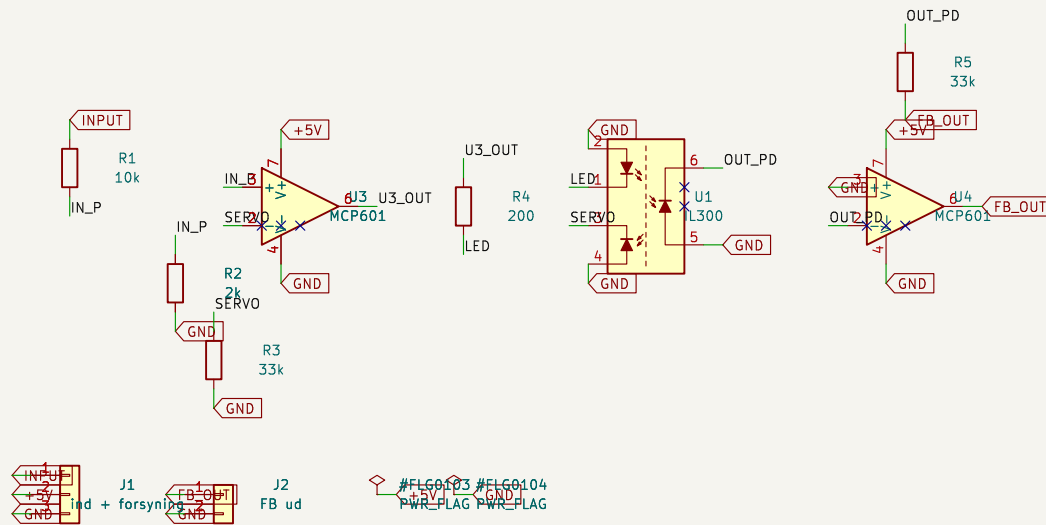
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Rev:
Id: 3/8

Feedback Circuit (Exp 3A: isolated linear amp – MCP601 → IL300 servo → MCP601)



Sheet: /Feedback/
File: feedback_circuit.kicad_sch

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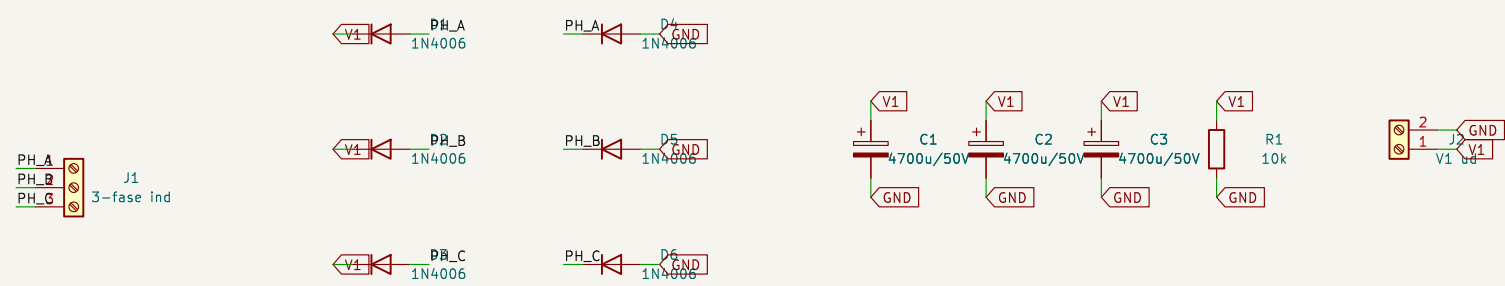
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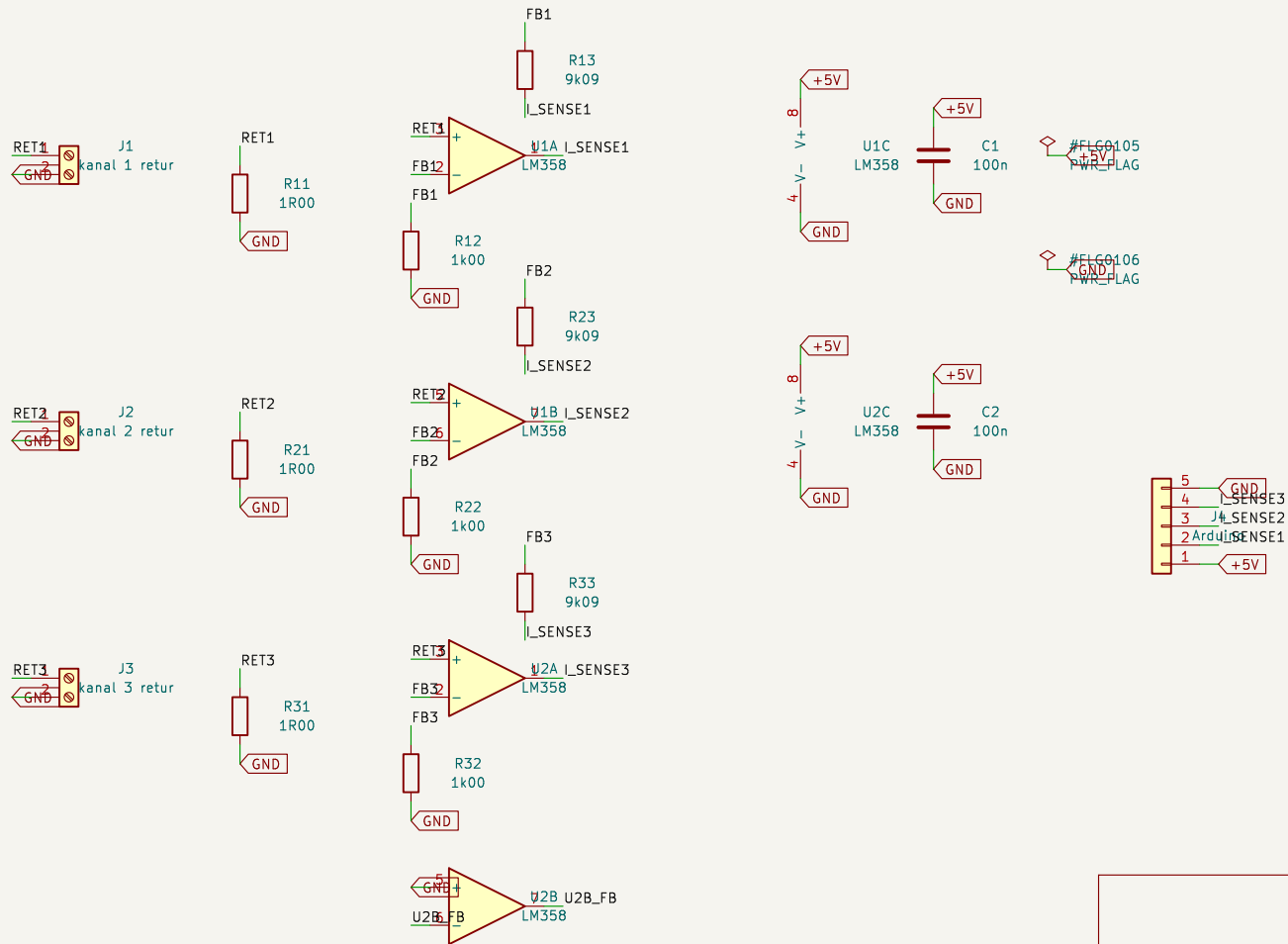
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Id: 4/8

3-fase ensretter + ladekondensator (gen -> 6x 1N4006 -> ~15 mF -> V1 15 V)



Stroemmaaling 3 kanaler (lavsidede 1R-shunt + LM358 x10 -> Arduino ADC)



Sheet: /Current sense/
File: current_sense.kicad_sch

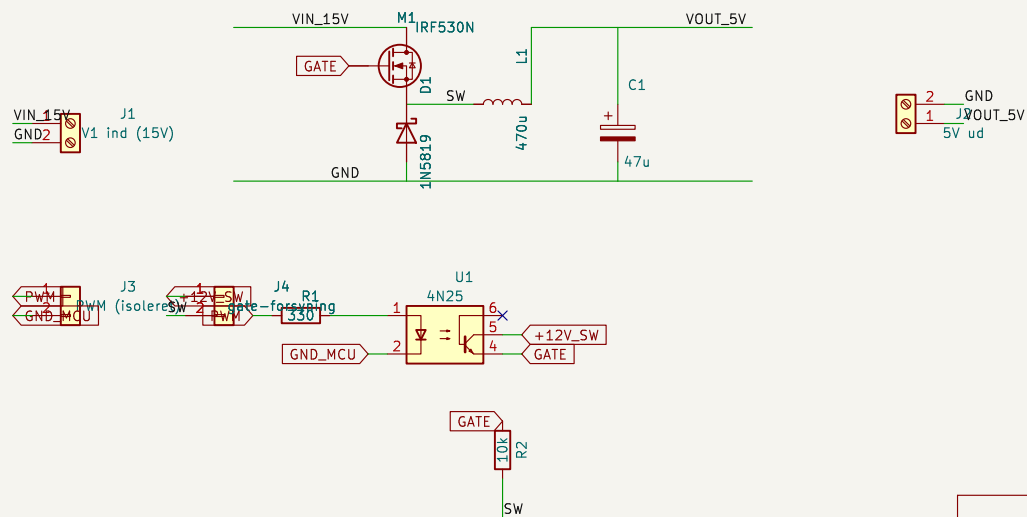
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Id: 6/8

Buck converter 15V → 5V (62768, Krav 8/9)



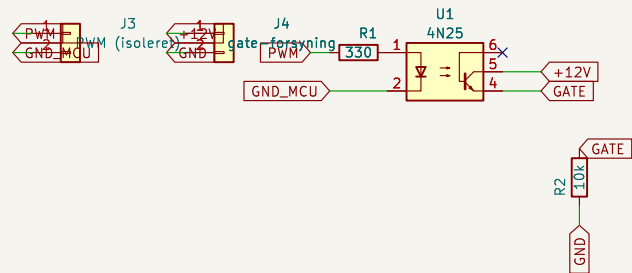
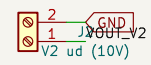
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